



• SOLAR ENERGY • LESSON 12

Inverter Sizing

for Solar PV Systems

Pure Sine • Continuous & Surge • Hybrid Inverters • Sizing Formula

What You Will Learn

- Compare pure sine, modified sine, and square wave outputs — and which loads need which
- Tell continuous (running) power apart from surge (starting) power — and why motors trip inverters
- Apply the inverter sizing formula with safety margins to size a unit correctly the first time
- Evaluate hybrid inverter-chargers and where they fit in off-grid and backup system design

Inverter Sizing Formula

$$P_{\text{cont}} \geq \Sigma P_{\text{running}} \times 1.25 \quad | \quad P_{\text{surge}} \geq P_{\text{largest_motor_surge}}$$

$\Sigma P_{\text{running}}$ = sum of loads that run at the same time | Always verify BOTH the continuous AND surge rating

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SOLAR-012: Inverter Sizing for Solar PV Systems

The inverter is the translator of every solar system — it converts the DC power produced by panels and stored in batteries into the AC power that household appliances, pumps, and tools actually use. Get the sizing wrong in one direction and the system trips every time a motor starts; get it wrong in the other direction and the client pays for capacity that sits idle, quietly wasting energy as standby losses every single day. Inverter sizing is a balance between "big enough to never trip" and "small enough to run efficiently."

Learning Objectives

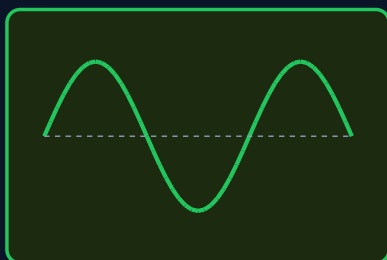
- Compare pure sine, modified sine, and square wave inverter outputs and identify which loads need which
- Distinguish continuous (running) power from surge (starting) power and explain why motors trip undersized inverters
- Apply the inverter sizing formula — with safety margins — to correctly size a unit for a real load list
- Read inverter efficiency curves and account for standby/idle consumption in a daily energy budget
- Evaluate hybrid inverter-chargers and identify where each type fits in an off-grid or backup design

1. Inverter Types and Output Waveforms

Solar inverters fall into three broad families: **grid-tie** (string) inverters that synchronise with and feed power into the utility grid and shut down when the grid fails; **off-grid (stand-alone)** inverters that create their own AC reference and run independently of the grid; and **hybrid** inverters that can do both — feeding the grid when it is present, running from battery when it is not, and charging the battery from either source. Whichever family is chosen, the shape of the AC waveform it produces decides which appliances will run safely on it.

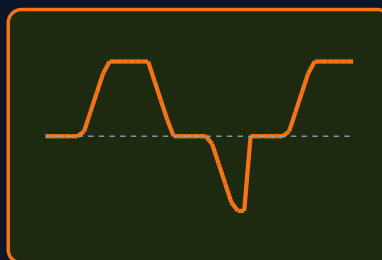
Inverter Output Waveforms — Pure Sine vs Modified Sine vs Square Wave

PURE SINE WAVE



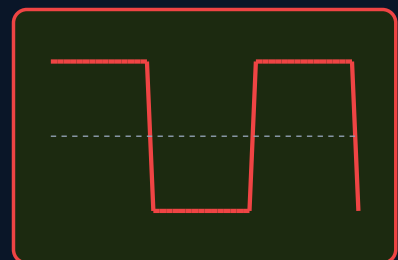
Matches utility power.
Safe for all loads —
motors, electronics,
medical equipment.

MODIFIED SINE WAVE



Stepped approximation.
Can cause motor hum,
timing errors in clocks
and microwaves. Cheaper.

SQUARE WAVE



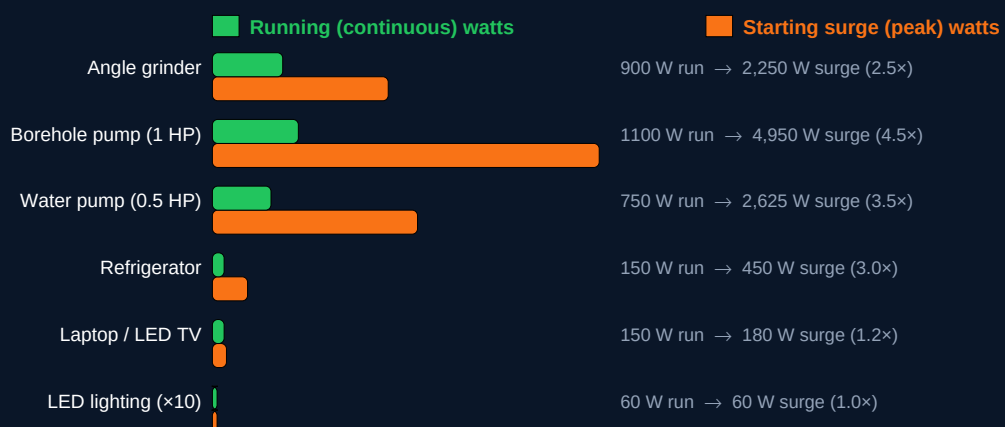
Crude on/off switching.
High harmonic content —
damages sensitive gear.
Obsolete for solar use.

Output Type	THD (Distortion)	Relative Cost	Compatible Loads	Typical Issues
Pure sine wave	< 3% — matches utility power	Highest	All loads — motors, compressors, electronics, medical equipment, audio/video gear, laser printers	None — behaves exactly like grid power
Modified sine wave	≈ 25–40%	Medium	Resistive loads (heaters, incandescent bulbs), basic chargers, simple power tools	Motor humming and overheating, dimmer flicker, microwave/clock timing errors, can damage sensitive electronics
Square wave	> 40%	Lowest (obsolete)	Very basic resistive loads only	Extremely high harmonic content damages most modern equipment — essentially obsolete for solar work

2. Continuous Power vs Surge (Starting) Power

Every appliance has two power ratings that matter for inverter sizing. **Continuous (running) power** is what the device draws once it is up and operating normally — this is the number printed on most nameplates. **Surge (starting/peak) power** is the brief, much higher current that motors, compressors, and pumps draw for one to three seconds at the moment they switch on, as the motor overcomes the inertia of a stationary rotor. An inverter that comfortably covers the running total can still trip instantly the moment a refrigerator compressor or borehole pump kicks in — because its surge rating, not its continuous rating, has been exceeded.

Running Watts vs Starting Surge — Why Motors Trip Undersized Inverters

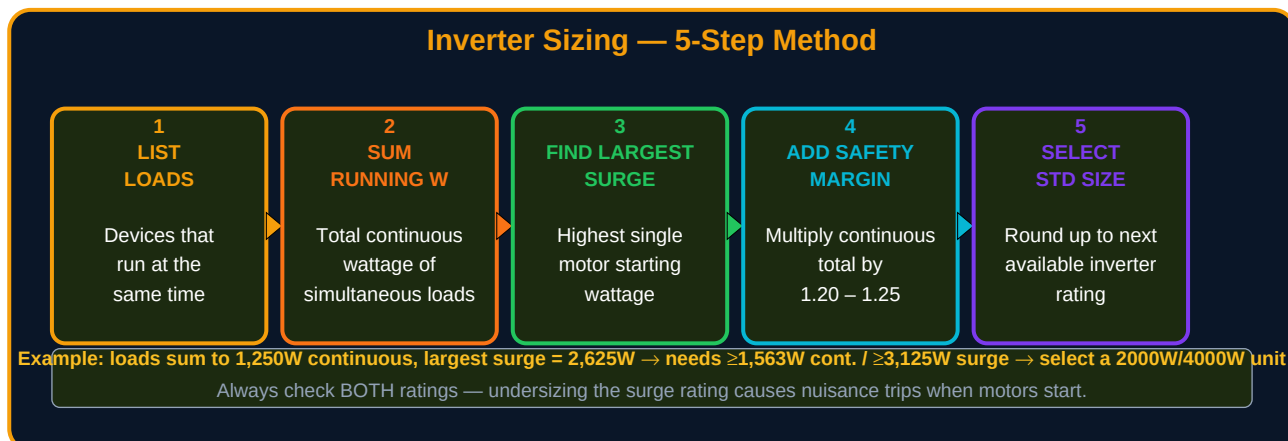


Notice that the borehole pump draws less running power than the angle grinder, yet its surge demand is by far the largest in the list. This is the single most common inverter sizing mistake:

engineers size for the sum of running watts and forget that one motor, starting alone, can momentarily demand more than the entire rest of the house combined.

Appliance / Load	Running W	Surge x	Surge W	Notes
LED lighting (×10 bulbs)	60 W	1.0×	60 W	No surge — solid-state, instant-on
Laptop / LED television	150 W	1.2×	180 W	Switch-mode power supply — small, brief inrush only
Refrigerator / freezer	150 W	3.0×	450 W	Compressor motor surges on every restart cycle, day and night
Water pump (0.5 HP)	750 W	3.5×	2,625 W	Single-phase induction motor — high locked-rotor starting current
Borehole / submersible pump (1 HP)	1,100 W	4.5×	4,950 W	Often the single largest surge in an off-grid system — frequently the number that sizes the inverter
Angle grinder / power tool	900 W	2.5×	2,250 W	Universal motor — moderate surge, very short duration

3. Inverter Sizing — Step by Step



Inverter Sizing Formula

$$P_{\text{continuous}} \geq \Sigma P_{\text{running}} \times 1.25 \text{ (safety margin)}$$

Where $\Sigma P_{\text{running}}$ is the sum of every load that may realistically run at the same time, and 1.25 is a typical safety margin (use 1.20–1.30 depending on how much load growth is expected). The continuous rating must always be checked alongside the surge rating below — sizing for one without the other produces a system that either trips on startup or wastes money on unused capacity.

$$P_{\text{surge}} \geq P_{\text{largest single motor surge}} + \Sigma P_{\text{other loads still running}}$$

Worked Example — Off-Grid House Inverter Sizing

Step	Calculation	Result
List simultaneous loads	Lights (200 W) + fridge (150 W run) + laptop (150 W) + water pump (750 W run, 0.5 HP) — realistically may all run together	4 loads, $\Sigma P_{\text{running}} = 1,250 \text{ W}$
Sum continuous (running) power	$\Sigma P_{\text{running}} = 200 + 150 + 150 + 750$	1,250 W continuous
Identify the largest single surge	Water pump surge = $750 \text{ W} \times 3.5 = 2,625 \text{ W}$. The other three loads (lights, fridge, laptop = 500 W) keep running at the same moment the pump starts.	Surge demand = $2,625 + 500 = 3,125 \text{ W}$
Apply the safety margin	$1,250 \text{ W} \times 1.25 \text{ safety factor}$	1,563 W minimum continuous rating
Select a standard inverter size	Round UP to the next available size that satisfies BOTH the continuous and surge requirement	2,000 W continuous / 4,000 W surge inverter ✓

4. Efficiency Curves and Standby Consumption

An inverter does not convert DC to AC for free — every watt that passes through it loses a small percentage to heat. That loss percentage is not constant: it changes with how heavily the inverter is loaded relative to its rated capacity. Manufacturers publish an efficiency curve for this exact reason, and reading it correctly is the difference between a system that sips battery power and one that quietly drains it around the clock.

Load Level	Typical Efficiency	What This Means
10% of rated capacity	70–80%	Significant losses relative to useful output — running an oversized inverter at very light loads wastes energy continuously
25% of rated capacity	85–90%	Acceptable — typical for early-morning and late-evening light-load periods
50% of rated capacity	92–96% (peak)	The sweet spot — most inverters are designed and rated to perform best in this range
75% of rated capacity	90–94%	Still efficient — a normal, healthy daytime operating range
100% of rated capacity	85–92%	Higher internal heating and losses — sustained full-load operation also shortens equipment lifespan

Beyond conversion losses, every off-grid and hybrid inverter also draws a small **standby (idle / search-mode) current** 24 hours a day just to stay switched on and ready to respond to a load. On a small system this can amount to several hundred watt-hours per day — energy that must be included in the daily energy budget calculated back in SOLAR-010, or the battery bank will be undersized without anyone realising why.

5. Hybrid Inverter-Chargers

A **hybrid inverter-charger** combines several functions that would otherwise be separate boxes — inverter, battery charger, automatic transfer switch, and sometimes even an MPPT solar charge controller — into a single unit. This simplifies wiring, reduces installation cost, and is now the standard approach for most residential off-grid and backup-power designs.

Model / Family	Type	Key Features	Typical Application
Victron MultiPlus / Quattro	Hybrid — inverter + charger + transfer switch	PowerAssist (blends grid and inverter power), seamless <20 ms transfer, VE.Bus parallel and 3-phase stacking	Off-grid homes, backup power, marine/RV, community mini-grids
SMA Sunny Island	Off-grid / hybrid battery inverter	Built for AC-coupled PV-plus-battery systems, multi-cluster operation (up to 9 units), frequency-shift power control	Larger off-grid systems and AC-coupled retrofits of existing grid-tie arrays
Growatt SPF / Deye SUN series	Hybrid — PV + battery + grid	Built-in MPPT solar charge controller, parallel operation, significantly lower cost than Victron or SMA	Budget-conscious residential and small commercial off-grid or hybrid systems
Generic all-in-one hybrid units	Hybrid all-in-one	Inverter, charger, MPPT controller, and battery management combined in a single enclosure	Compact residential systems where installation space and simplicity matter most

6. Practice Exercises

#	Exercise	Task
1	Waveform selection	A client wants to power a chest freezer, a laptop, and an old CRT television from an off-grid inverter. Which output waveform must you specify, and what would go wrong with each device if you chose the cheaper modified-sine option instead?
2	Surge identification	A home runs these loads together: lights (300 W), fridge (150 W run / 3.0× surge), laptop (100 W), and a 1 HP borehole pump (1,100 W run / 4.5× surge). Calculate $\Sigma P_{\text{running}}$ and the worst-case surge demand at the moment the pump starts.
3	Full inverter sizing	Using the load list from Exercise 2: (a) apply a 1.25 safety margin to size the continuous rating, (b) determine the minimum surge rating, (c) select a standard inverter size that satisfies both ratings.

4	Efficiency analysis	A homeowner installs a 5,000 W inverter but typically only runs about 600 W of loads (12% of rated capacity). Using the efficiency table in Section 4, explain what is happening to the "missing" energy, and what a better-sized choice would have looked like.
5	Hybrid system design	A small rural clinic needs grid backup, battery storage, and solar charging in one compact system with very limited installation space. Recommend an inverter category from Section 5 and justify your choice against at least one alternative.

Lesson Summary

Key Concept	Summary Rule
Pure sine wave, always	Specify pure sine wave inverters for solar work — modified sine wave causes problems with motors and modern electronics, making it a false economy.
Continuous $\neq$ surge	Running watts and starting watts are different numbers. Motors and compressors can surge 3–5 $\times$ their running wattage for one to three seconds at startup.
The sizing rule	$P_{\text{continuous}} \geq \Sigma P_{\text{running}} \times 1.25$ safety margin. $P_{\text{surge}} \geq$ largest single motor surge + the other loads still running at that instant.
One motor can size the whole system	In most off-grid designs, the single largest motor surge — not the sum of running loads — is what determines the minimum inverter surge rating.
Efficiency has a sweet spot	Inverters run most efficiently at roughly 40–75% of rated capacity. An oversized unit running mostly at 10–20% load wastes energy continuously, every single day.
Standby losses are real	Off-grid and hybrid inverters draw idle/search-mode power 24 hours a day — include this in the daily energy budget, especially on small systems.
Hybrid inverter-chargers simplify design	Combining inverter, charger, transfer switch, and sometimes MPPT into one unit is now the standard approach for residential off-grid and backup systems.

What Is Next — SOLAR-013

SOLAR-013: Solar Panel Sizing — How to translate a daily energy budget into an actual array: panel wattage and efficiency, peak sun hours and seasonal variation, string sizing and MPPT voltage windows, derating factors (temperature, soiling, wiring losses), and a complete worked example sizing a panel array to match the battery bank designed in SOLAR-011.

Resource	Details
SOLAR PDF Series	SOLAR-001 through SOLAR-030 — ayetechub.com/pdfs.html
Solar PV Calculator	Free interactive tool — ayetechub.com/downloads/solar-calculator.html
Community	Telegram: t.me/ayetechub

Quick Glossary — Terms Used in This Lesson

Term	Meaning
Continuous (running) power	The steady wattage a device draws once it is up and operating normally — the number printed on its nameplate.
Surge (starting/peak) power	The brief, much higher current a motor, compressor, or pump draws for one to three seconds at the moment it switches on.
Pure sine wave	An AC output that matches utility power exactly — the only waveform that runs every load safely, including motors and sensitive electronics.
MPPT	Maximum Power Point Tracking — the charge-controller technology built into many hybrid inverters that extracts the most possible power from a solar array.
Standby (search-mode) current	The small amount of power an inverter draws around the clock just to stay switched on and ready to respond to a load.